

# PATHOLOGY REPORT

ARISE • Autonomous Multi-Agent Diagnosis Assistant — Demo Input

|            |                           |               |                |
|------------|---------------------------|---------------|----------------|
| Patient ID | SYN-PT-0427               | Report ID     | PATH-2026-0427 |
| Age / Sex  | 56 / Female               | Specimen Date | 24 Sep 2026    |
| Specimen   | Core biopsy — breast mass | Status        | Final report   |

## CLINICAL HISTORY

Patient with a palpable mass in the upper outer quadrant of the left breast. Imaging demonstrated an irregular solid lesion measuring approximately 2.1 cm. Core biopsy submitted for histopathological evaluation.

## GROSS DESCRIPTION

Multiple tan-white, cylindrical tissue fragments are received in formalin, measuring 0.3–1.4 cm in greatest dimension. Entire specimen submitted for processing.

## MICROSCOPIC DESCRIPTION

Sections show breast tissue infiltrated by irregular nests and small glands of atypical epithelial cells within a desmoplastic stromal background. The cells demonstrate enlarged hyperchromatic nuclei, moderate pleomorphism, and occasional mitotic figures. Focal lymphovascular spaces are suspicious for involvement. No definite in-situ component is identified in the sampled tissue.

## IMMUNOHISTOCHEMISTRY

Tumor cells show strong membranous staining for E-cadherin. Estrogen receptor (ER) shows diffuse nuclear positivity in the majority of tumor cells. Progesterone receptor (PR) shows moderate nuclear positivity. HER2 immunohistochemistry is reported as equivocal (2+); reflex in-situ hybridization is recommended for final HER2 assessment.

## PATHOLOGY IMPRESSION

**Invasive carcinoma of breast, consistent with invasive ductal carcinoma of no special type (NST), based on the sampled tissue.**

## RECOMMENDATION / COMMENT

Correlation with imaging and clinical findings is recommended. Additional biomarker testing, including reflex HER2 in-situ hybridization where applicable, should be interpreted with the complete clinical record. The findings in this report are intended for clinician review and should not be used as a standalone diagnosis.

**DEMO DATA NOTICE:** This is a synthetic fictional pathology report created for the ARISE hackathon prototype. It does not represent a real patient or a real clinical report.